

Notes: Barkai (JF, 2020)
Declining Labor and Capital Shares

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1 Big picture

1.1 Main question

The paper asks what replaced the lost labor share in U.S. corporate value added. A common story says that firms substituted capital for labor: robots, software, equipment, and other capital inputs became cheaper, firms used more capital, and capital owners captured a larger share of output. Barkai’s core question is whether this substitution story is visible in the data once the cost of using capital is measured directly.

1.2 Core answer

The answer is no. The labor share falls, but the capital share also falls. The missing share is accounted for by a large increase in pure profits. In Barkai’s interpretation, this is evidence that the decline in the labor share is more naturally connected to rising market power and declining competition than to a simple substitution from labor toward capital.

1.3 Why this paper matters

The paper changes the accounting of the labor-share debate. Many papers define the nonlabor share as “capital’s share,” but that is only correct if pure profits are zero. Barkai separates the nonlabor share into two pieces:

1. **capital costs**: the required return on the capital stock, and
2. **pure profits**: residual earnings after paying labor, capital, and indirect taxes.

This distinction matters because a shift from labor to capital has very different economic and policy implications than a shift from labor and capital to pure profits. The first points to technology and input substitution; the second points to market power, weak competition, entry barriers, common ownership, regulation, or other sources of rents.

2 Incremental contribution of the paper

2.1 Directly measures capital costs instead of assuming zero profits

The central contribution is measurement. The paper constructs capital costs as

$$E_t = \sum_s R_{s,t} P_{s,t}^K K_{s,t},$$

where $R_{s,t}$ is the required rate of return on asset type s and $P_{s,t}^K K_{s,t}$ is the nominal value of the asset stock. This is conceptually analogous to measuring a firm’s annual leasing costs for every type of capital input. The innovation is that Barkai does this even when firms own rather than lease the capital.

2.2 Separates capital costs from pure profits

The paper emphasizes that the residual after paying labor is not automatically a payment to capital. Pure profits are constructed as

$$\Pi_t = P_t^Y Y_t - w_t L_t - R_t P_t^K K_t - \text{indirect taxes}_t.$$

The pure profit share is $S_t^\Pi = \Pi_t / (P_t^Y Y_t)$, while the capital share is $S_t^K = R_t P_t^K K_t / (P_t^Y Y_t)$. This separation is the paper's key accounting move.

2.3 Uses market prices to measure the required return on capital

The required return is not assumed constant. It is constructed from the cost of debt, the cost of equity, expected capital inflation, depreciation, and tax treatment. Because risk-free rates and debt/equity costs decline substantially after the early 1980s, the required return on capital falls sharply. This is exactly why it is important not to assume a fixed user cost of capital.

2.4 Shows the capital share declines, not rises

A standard capital-labor substitution view predicts that cheaper capital should lead firms to use enough additional capital that capital payments rise relative to value added. Barkai finds the opposite: the required return falls, and the capital stock does not rise enough to offset it. The capital share declines from about 32% to 25% of gross value added.

2.5 Connects the aggregate evidence to industry concentration

The paper supplements aggregate accounting with cross-industry regressions using Economic Census data. Industries where concentration rises more also tend to experience larger labor-share declines. This provides cross-sectional evidence that supports the market-power interpretation and avoids relying directly on capital-stock measurement.

3 Core accounting framework

3.1 Capital costs by asset

For each asset type s , capital costs are

$$E_s = R_s P_s^K K_s.$$

Here $P_s^K K_s$ is the value of the capital stock and R_s is the annual required return. Barkai follows the Hall-Jorgenson user-cost tradition. A simplified version is $R_s \approx r + \delta_s - \mathbb{E}[\pi_s]$, but the paper's actual measure also incorporates debt and equity financing and tax treatment.

3.2 Required rate of return

The asset-specific required return is

$$R_s = \left(\frac{D}{D+E} i^D (1-\tau) + \frac{E}{D+E} i^E \right) - \mathbb{E}[\pi_s] + \delta_s \quad \text{with tax adjustment factors.}$$

The first term is the weighted average cost of capital. Expected capital inflation lowers the user cost because holding capital becomes cheaper when its price is expected to rise. Depreciation raises the user cost because worn-out capital must be replaced.

3.3 Aggregate shares

Aggregate capital costs are summed across assets:

$$E = \sum_s R_s P_s^K K_s.$$

The capital share is

$$S^K = \frac{\sum_s R_s P_s^K K_s}{P^Y Y}.$$

The pure profit share is

$$S^\Pi = \frac{P^Y Y - wL - RP^K K - \text{indirect taxes}}{P^Y Y}.$$

The labor share is compensation of employees divided by gross value added. The three shares, together with indirect taxes, exhaust gross value added.

4 Data and measurement choices

4.1 Main aggregate data

The main analysis uses the U.S. nonfinancial corporate sector from 1984 to 2014. This sector excludes financial firms and focuses on corporate domestic production. Gross value added, employee compensation, and taxes come from NIPA. Capital stocks and depreciation come from BEA fixed asset tables. The main specification includes BEA fixed assets and inventories.

4.2 Debt cost of capital

The debt cost of capital is a market-value-weighted average of bond yields, commercial paper rates, and bank loan rates. Because some detailed rate series do not span the full sample, Barkai validates longer available proxy series against shorter more detailed measures. This is the purpose of Figure 1.

4.3 Equity cost of capital

The baseline equity cost of capital equals the 10-year Treasury yield plus a constant 5% equity premium. Barkai argues that the literature tends to find either a constant or declining equity risk

premium since the 1980s. If the true equity premium declined, then the baseline understates the rise in pure profits; if it increased, the baseline overstates the rise. Figure 5 studies robustness to this issue.

4.4 Capital inflation and depreciation

Expected capital inflation is constructed as a three-year moving average of realized capital inflation. Depreciation rates are taken from BEA fixed asset tables. Barkai studies robustness to alternative depreciation and capital inflation assumptions in Figure 4.

5 Main quantitative findings

5.1 Required return on capital declines substantially

The weighted average cost of capital declines from roughly 11.3% in 1984 to 6.3% in 2014. The required rate of return on capital declines from roughly 18.9% to 13.3%, a 5.6 percentage point drop. The decline in the required return is primarily driven by the decline in the cost of capital rather than by changes in depreciation or inflation.

5.2 Capital share falls

The capital share declines from roughly 32% of gross value added in 1984 to roughly 25% in 2014. This is a 7 percentage point decline, or about 22% relative to the initial level. Importantly, this decline is larger in percentage terms than the 11% decline in the labor share.

5.3 Pure profit share rises sharply

The pure profit share increases from about -5.6% of gross value added in 1984 to about 7.9% in 2014, an increase of 13.5 percentage points. Barkai reports that this amounts to more than \$1.2 trillion in 2014, or about \$14,600 per employee in the nonfinancial corporate sector.

5.4 Labor has not been replaced by capital

A useful way to summarize the finding is that every dollar of labor costs was accompanied by about 49 cents of capital costs in 1984 but only 42 cents in 2014. Thus, despite the decline in the labor share, labor costs grew faster than capital costs. The evidence is inconsistent with the claim that the aggregate decline in labor share was simply a shift from labor payments to capital payments.

6 Industry-concentration evidence in detail

6.1 Empirical design

The industry analysis uses Economic Census data for 1997, 2002, 2007, and 2012. The unit of observation is a six-digit NAICS industry. Concentration is measured as the sales share of the largest 4, 8, 20, and 50 firms. The labor share is payroll divided by sales. Barkai estimates first-difference and log-difference regressions of changes in the labor share on changes in concentration, weighting industries by sales and clustering standard errors by three-digit NAICS industry.

6.2 Interpretation of the regression evidence

The industry evidence is not a randomized causal design. It is best understood as a reduced-form correlation that is consistent with the aggregate pure-profit evidence. Industries becoming more concentrated are industries where labor receives a smaller share of sales. This is what many imperfect-competition models predict: as market power rises, firms restrict input demand relative to output, and the shares paid to labor and capital can fall while markups and profits rise.

6.3 Why tradable industries matter

The tradable-industry split is useful because Census concentration may overstate true product-market concentration when foreign competitors matter. If domestic concentration rises but imports discipline prices, domestic concentration is a weaker proxy for market power. Barkai finds the concentration-labor-share relationship is much stronger outside tradable industries, which strengthens the market-power interpretation.

7 Figures and tables

7.1 Figure 1: validating debt-cost proxies

Figure 1 is not a main result about labor or capital shares; it is a data validation figure. Panel A compares Barclays bond-index yields to Moody's Aaa and Baa yields. The key point is that Moody's Aaa tracks the Barclays aggregate closely in the overlapping period, so it can serve as a longer historical proxy. Panel B similarly validates the 3-month AA nonfinancial commercial paper rate against a more detailed aggregate commercial paper series. Panel C validates the prime lending rate as a proxy for commercial and industrial loan rates before the detailed loan data begin. The interpretation is that Barkai's debt cost of capital is not built on arbitrary proxies; the proxies are chosen because they match detailed data in overlap periods.

7.2 Figure 2: why the required return falls

Figure 2 decomposes the required return on capital. Panel A shows that debt, equity, and WACC all decline over time. Panel B shows expected inflation is roughly trendless, so the user cost decline is not driven by inflation assumptions. Panel C shows depreciation is fairly stable. Panel D puts these components together and shows a large downward trend in the required return. This figure is the mechanical core of the paper: if the required return falls and capital accumulation is not sufficiently strong, the capital share must fall.

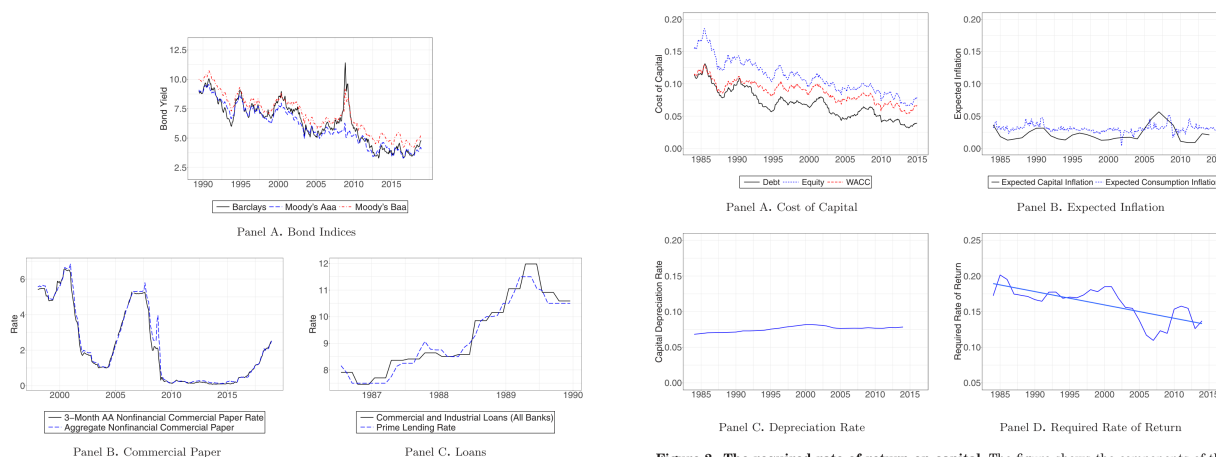


Figure 1. Bond indices, commercial paper, and loans. Panel A (Bond Indices) shows the aggregate option-adjusted yield on the Barclays indices and Moody's Aaa and Baa Corporate Bond Yields. Panel B (Commercial Paper) shows the aggregate rate of the 12 types of nonfinancial commercial paper and the 3-Month AA Nonfinancial Commercial Paper Rate. Panel C (Loans) shows the aggregate rate on commercial and industrial loans made by all commercial banks and the

Figure 1: Debt-cost proxy validation. The three panels show that longer proxy series track shorter detailed debt-rate measures closely.

Figure 2. The required rate of return on capital. The figure shows the components of the required rate of return on capital for the U.S. nonfinancial corporate sector over the period 1984 to 2014. Panel A: the debt cost of capital, the equity cost of capital, and the weighted average cost of capital. Panel B: expected capital inflation is calculated as a three-year moving average of realized capital inflation and expected consumption inflation is the median expected 12-month price change from the University of Michigan's Survey of Consumers. Panel C: the depreciation rate of capital is taken from the BEA Fixed Asset Tables. Panel D: the required rate of return on capital with a linear trend. See Section III for further details. (Color figure can be viewed at wileyonlinelibrary.com)

Figure 2: Required return decomposition. The decline in WACC drives the decline in the required rate of return.

7.3 Figure 3: the central result

Figure 3 is the paper's main visual result. Panel A shows that the capital share declines; Panel B shows that the pure profit share rises. The two panels should be read together. The decline in

labor's share does not show up as an increase in capital's share. Instead, it shows up as an increase in pure profits. This is the visual evidence behind the paper's claim that rising profits, not capital deepening, offset the decline in labor's share.

7.4 Figure 4: robustness to depreciation and capital inflation

Figure 4 asks whether the increase in pure profits is an artifact of depreciation or capital inflation assumptions. Panel A shows almost no sensitivity to capital-inflation adjustments. Panel B shows modest variation when all depreciation rates are multiplied up or down. Panel C shows larger sensitivity when depreciation has a trend, but even over a wide range the increase in the pure profit share remains large, roughly 12 to 15 percentage points. The interpretation is that reasonable measurement errors in depreciation and capital inflation cannot eliminate the main finding.

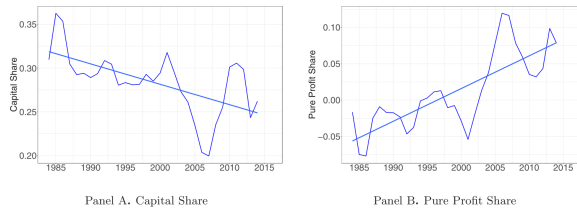


Figure 3. Capital and pure profit shares. The figure shows the capital share and pure profit share of gross value added for the U.S. nonfinancial corporate sector over the period 1984 to 2014. Capital costs are the product of the required rate of return on capital and the value of the capital stock. Pure profits are gross value added less compensation of employees less capital costs less taxes on production and imports plus subsidies. Panel A: the capital share is the ratio of capital costs to gross value added. Panel B: the pure profit share is the ratio of pure profits to gross value added. Both figures include a fitted linear trend. See Section III for further details. (Color figure can be viewed at wileyonlinelibrary.com)

Figure 3: Main result. Capital share falls while pure profit share rises.

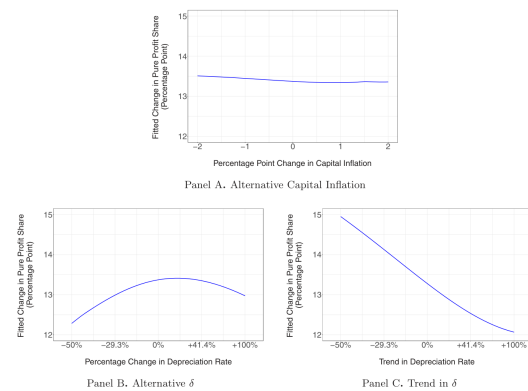


Figure 4. Alternative values of capital inflation and the rate of depreciation. The figure shows the fitted change in the pure profit share for a range of adjustments to capital inflation and the rate of depreciation. Panel A: adjustments to capital inflation. Panel B: constant multiplicative adjustments to the rate of depreciation. Panel C: adjustments to the trend in the rate of depreciation. See Section IV A for further details. (Color figure can be viewed at wileyonlinelibrary.com)

Figure 4: Robustness. Pure-profit trend remains large under alternative inflation and depreciation assumptions.

7.5 Table I: alternative capital measures

Table I shows that the results are not driven by a narrow definition of capital. The main specification, which includes BEA capital and inventories, gives an 11% decline in the labor share, a 22% decline in the capital share, and a 13.5 percentage point increase in the pure profit share. Excluding inventories still gives a large pure-profit increase of 11.5 percentage points. Adding real estate valued at market prices makes the capital-share decline larger and the pure-profit increase larger. This is important because adding more measured capital does not automatically rescue the capital-substitution story: when the required return falls, additional capital must grow very fast relative to output to prevent the capital share from declining.

7.6 Table II: missing intangible capital scenarios

Table II lays out the scenario analysis for omitted intangible capital. It defines hypothetical investment, inflation, depreciation, and initial-stock assumptions. The table also benchmarks

these assumptions against existing estimates of economic competencies and IPP capital. The key interpretation is that omitted intangible capital can reduce measured pure profits only if it is very large, grows fast, and has a low enough depreciation rate. The actual fitted values for economic competencies and IPP have relatively high depreciation rates, especially economic competencies, which makes them less capable of explaining the persistent rise in pure profits.

Table I
Time Trends of Labor, Capital, and Pure Profit Shares

The table reports time trends for the U.S. nonfinancial corporate sector over the period 1984 to 2014. This table considers alternative measures of capital. BEA capital data are taken from BEA Fixed Asset Table 4.1 and include structures, equipment, and intellectual property products. Data on inventories and real estate valued at market prices are taken from the Integrated Macroeconomic Accounts for the United States, Table S.5.a. Capital data in column (1) consist of BEA capital and inventories. Capital data in column (2) consist of BEA capital. Capital data in column (3) consist of BEA capital and inventories and real estate is valued at market prices. Capital costs are the product of the required rate of return on capital and the value of the capital stock. Expected capital inflation is calculated as a three-year moving average of realized capital inflation. Pure profits are gross value added less compensation of employees less capital costs less taxes on production and imports plus subsidies. The labor share is the ratio of compensation of employees to gross value added. The capital share is the ratio of capital costs to gross value added. The pure profit share is the ratio of pure profits to gross value added. For a variable X , the fitted percentage point (pp) change in X is $X_{2014} - X_{1984}$, and the fitted percent (%) change in X is $\frac{X_{2014} - X_{1984}}{X_{1984}}$. The increase in pure profits per employee is the fitted percentage point change in the pure profit share multiplied by gross value added in 2014 and divided by the number of employees in 2014. See Sections III and IV.B for further details.

	Measure of Capital		
	(1 – Main)	(2)	(3)
Decline in Labor Share	11%	11%	11%
Decline in Capital Share	22%	17%	28%
Increase in Pure Profit Share	13.5pp	11.5pp	16.2pp
Increase in Pure Profits Per Employee	\$14.6 (thousand)	\$12.4 (thousand)	\$17.5 (thousand)

- (1) BEA Measures of Capital and Inventories.
(2) BEA Measures of Capital.
(3) BEA Measures of Capital, Inventories, and Real Estate Valued at Market Prices.

Table I: Alternative capital definitions. More inclusive capital measures do not eliminate the pure-profit increase.

7.7 Figure 5: robustness to the equity risk premium

Figure 5 asks whether the rise in pure profits depends on the assumed equity premium. Panel A shows that higher constant equity premia reduce the measured increase in pure profits, because a higher equity cost raises capital costs. But even with an 8% equity risk premium, the pure profit share still rises substantially. Panel B allows the equity premium to trend. A rising equity premium dampens the profit increase; a falling premium magnifies it. The main takeaway is that only a large upward trend in the equity premium would materially weaken the result, while much of the asset-pricing literature points toward a flat or declining premium.

7.8 Figure 6: why constant-user-cost measures miss the decline

Figure 6 plots investment divided by gross value added. It does not show a clear decline over 1984–2014. This matters because some earlier capital-share measures effectively infer the capital share from investment shares under assumptions of a constant required return and a stable capital-to-investment ratio. Barkai argues that such methods miss the decline in the capital share because the key object that falls is the required return on capital, not investment relative to value added.

7.9 Table III: industry-level facts

Table III reports descriptive statistics for 750 six-digit NAICS industries. The median labor share of sales declines from 19.80 in 1997 to 17.70 in 2012, while the mean labor share declines from 21.47

Table II
Parameter Inputs into Missing Capital Scenarios

The table reports the functional form assumptions and the range of parameter values for the construction of unmeasured intangible capital. P^X is nominal investment in unmeasured intangible capital. π^X is inflation of unmeasured intangible capital. δ^X is the rate of depreciation of unmeasured intangible capital. $P^Y X$ is the nominal value of unmeasured intangible capital. $P^Y Y$ is measured gross value added and π^Y is the percent change in the price deflator of gross value added for the nonfinancial corporate sector (taken from NIPA Table 1.14). The nominal value of unmeasured capital at the end of period t is constructed by the perpetual inventory method and is given by the equation $P^X X_{t+1} = (1 - \delta^X) P^X X_t + P^Y Y_t$. Data on Economic Competencies are taken from IntanInvest. Data on Intellectual Property Products are taken from the BEA Fixed Asset Table 4.1. See Section IV.C for further details.

Input	Assumed Form	Range of Values	Fitted Values	
			Economic Competencies	Intellectual Property Products
Investment	$\frac{P^X}{P^Y} X_t = a + b \times (t - 1984)$	$a \in [0, 0.2]$	$a = 0.067$	$a = 0.036$
Inflation	$\pi_t^X = \pi_t^Y + c$	$b \in [0, 0.01]$	$b = 0.0005$	$b = 0.0009$
Depreciation	$\delta^X = d$	$c \in [-0.02, 0.02]$	$c = 0.008$	$c = 0.002$
Initial Stock	$P_{1979}^X X_{1979} = e \times P_{1979}^Y Y_{1979}$	$d \in [0.05, 0.6]$	$d = 0.44$	$d = 0.22$
		$e \in [0, 2]$		$e = 0.117$

Table II: Missing-capital scenario inputs. Omitted intangibles would need to be large and persistent to explain away profits.

to 20.28. Concentration rises across all measures: for example, the mean sales share of the four largest firms rises from 30.57 to 35.85, and the median rises from 25.95 to 32.50. The table shows that the industry-level exercise starts from two broad facts: labor shares fall and concentration rises within the same period.

7.10 Table IV: concentration predicts labor-share declines

Table IV is the main first-difference regression. The dependent variable is the change in an industry’s labor share. The key coefficient is negative for all concentration measures. For the four-firm concentration ratio, the coefficient is -0.113 , statistically significant at 1%. This means that a 10 percentage point increase in the sales share of the largest four firms is associated with roughly a 1.13 percentage point decline in the labor share. The predicted decline from the concentration change is about as large as the observed decline in the first-difference specification. This is strong reduced-form evidence that rising concentration is closely connected to falling labor shares.

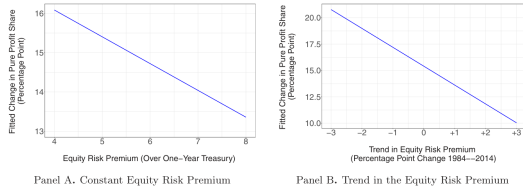


Figure 5. Alternative values of the equity risk premium. The figure shows the fitted change in the pure profit share for a range of adjustments to the equity risk premium. The equity cost of capital is the sum of the yield on a one-year U.S. treasury and the equity risk premium. Panel A: adjustments to the level of the equity risk premium. This panel assumes a constant equity risk premium. Panel B: adjustments to the trend in the equity risk premium. This panel calculates the equity risk premium in year t as $5\% + \text{Trend} \times \frac{t-1984}{2014-1984}$. See Section IV.D for further details. (Color figure can be viewed at wileyonlinelibrary.com.)

Figure 5: Equity-risk-premium robustness. The profit trend remains positive under a wide range of equity-premium assumptions.

**Table III
Descriptive Statistics**

The table reports descriptive statistics of the matched sample of census industries. Data on industry payroll, sales, and concentration are taken from the economic census. The unit of observation is a six-digit NAICS industry. See Section V for further details.

	<i>N</i>	Median	Mean	<i>SD</i>
Value in 1997				
Labor Share	750	19.80	21.47	11.87
Sales Share of 4 Largest Firms	748	25.95	30.57	20.87
Sales Share of 8 Largest Firms	747	37.40	40.09	24.62
Sales Share of 20 Largest Firms	750	52.15	52.13	27.31
Sales Share of 50 Largest Firms	749	67.00	63.02	27.85
Value in 2012				
Labor Share	750	17.70	20.28	12.88
Sales Share of 4 Largest Firms	748	32.50	35.85	21.78
Sales Share of 8 Largest Firms	747	44.10	45.86	24.72
Sales Share of 20 Largest Firms	750	60.05	57.84	26.47
Sales Share of 50 Largest Firms	749	75.50	68.22	26.42
Change in Value (1997 to 2012)				
Labor Share	750	-1.41	-1.19	5.90
Sales Share of 4 Largest Firms	748	4.15	5.28	12.10
Sales Share of 8 Largest Firms	747	4.70	5.77	11.80
Sales Share of 20 Largest Firms	750	4.10	5.71	10.93
Sales Share of 50 Largest Firms	749	3.20	5.20	9.86
Log-Change in Value (1997 to 2012)				
Labor Share	750	-0.08	-0.10	0.28
Sales Share of 4 Largest Firms	748	0.17	0.21	0.46
Sales Share of 8 Largest Firms	747	0.13	0.18	0.38
Sales Share of 20 Largest Firms	750	0.09	0.14	0.30
Sales Share of 50 Largest Firms	749	0.05	0.11	0.23

Table III: Industry facts. Labor shares fall and concentration measures rise from 1997 to 2012.

Table IV
Labor Share and Industry Concentration—Regression in First Differences

The table reports results of regressions of changes in the labor share on changes in industry concentration. The unit of observation is a six-digit NAICS industry. Observations are weighted by an industry's share of sales. Standard errors are clustered by three-digit NAICS industry. Data on industry payrolls, sales, and concentration are taken from the economic census. The observed decline is the sales-weighted average change in the labor share. The predicted decline is the sales-weighted average predicted change in the labor share. See Section V for further details. ^{***}, ^{**}, and ^{*} denote significance at the 1%, 5%, and 10% levels, respectively.

	Dependent Variable: $S_{jt}^L - S_{jt-5}^L$			
	(1)	(2)	(3)	(4)
$C_{jt}^{(4)} - C_{jt-5}^{(4)}$	-0.113 ^{***} (0.029)			
$C_{jt}^{(8)} - C_{jt-5}^{(8)}$		-0.108 ^{***} (0.028)		
$C_{jt}^{(20)} - C_{jt-5}^{(20)}$			-0.125 ^{***} (0.031)	
$C_{jt}^{(50)} - C_{jt-5}^{(50)}$				-0.133 ^{**} (0.036)
Year FE	Yes	Yes	Yes	Yes
R^2 (Within)	0.07	0.06	0.07	0.06
Observations	2,224	2,232	2,229	2,235
Observed decline	-0.81	-0.84	-0.81	-0.80
Predicted decline	-0.84	-0.98	-1.25	-1.24

Table IV: First-difference regressions. Concentration increases predict labor-share declines.

Table V
Labor Share and Industry Concentration—Regression in Log Differences

The table reports results of regressions of log-changes in the labor share on log-changes in industry concentration. The unit of observation is a six-digit industry. Observations are weighted by an industry's share of sales. Standard errors are clustered by three-digit NAICS industry. Data on industry payrolls, sales, and concentration are taken from the economic census. The observed decline is the sales-weighted average change in the log-labor share. The predicted decline is the sales-weighted average change in the predicted change in the log-labor share. See Section V for further details. ^{***}, ^{**}, and ^{*} denote significance at the 1%, 5%, and 10% levels, respectively.

	Dependent Variable: $\log S_{jt}^L - \log S_{jt-5}^L$			
	(1)	(2)	(3)	(4)
$\log C_{jt}^{(4)} - \log C_{jt-5}^{(4)}$	-0.215 ^{***} (0.079)			
$\log C_{jt}^{(8)} - \log C_{jt-5}^{(8)}$		-0.242 ^{**} (0.110)		
$\log C_{jt}^{(20)} - \log C_{jt-5}^{(20)}$			-0.318 ^{**} (0.151)	
$\log C_{jt}^{(50)} - \log C_{jt-5}^{(50)}$				-0.424 ^{**} (0.197)
Year FE	Yes	Yes	Yes	Yes
R^2 (Within)	0.07	0.06	0.06	0.06
Observations	2,224	2,232	2,229	2,235
Observed decline	-0.22	-0.22	-0.22	-0.22
Predicted decline	-0.07	-0.08	-0.08	-0.09

Table V: Log-difference regressions. The concentration relation remains negative and significant in log changes.

7.11 Table V: log-difference specification

Table V repeats the concentration-labor-share regression in logs. The coefficients remain negative and statistically significant. In this specification, the predicted decline is smaller than in Table IV—roughly one-third to 40% of the observed log decline. This makes the evidence more conservative: the sign and significance survive, but the magnitude is less complete than in levels. The table helps show that the relation is not an artifact of using percentage-point changes.

7.12 Table VI: subsample robustness

Table VI asks whether the concentration result is driven by specific sectors. Excluding tradable industries produces an even more negative coefficient (-0.131) and a predicted decline larger than the observed decline. In tradable industries alone, the coefficient is small and only weakly significant. Excluding sectors with tax-exempt firms or excluding R&D-intensive industries does not remove the result. The interpretation is that the concentration-labor-share relation is strongest where domestic concentration is more likely to measure product-market power, and it is not simply a story about R&D-intensive or tax-exempt sectors.

Table VI
Labor Share and Industry Concentration—By Subsample

The table reports results of regressions of changes in the labor share on changes in industry concentration. The unit of observation is a six-digit NAICS industry. Observations are weighted by an industry's share of sales. Standard errors are clustered by three-digit NAICS industry. Data on industry payrolls, sales, and concentration are taken from the economic census. The observed decline is the sales-weighted average change in the labor share. The predicted decline is the sales-weighted average predicted change in the labor share. The classification of tradable industries is taken from Mian and Sufi (2014). Column (4) excludes Health Care and Social Assistance (NAICS 62) and Other Services (NAICS 81). The classification on R&D industries is based on the NSF R&D survey. See Section V for further details. ^{***}, ^{**}, and ^{*} denote significance at the 1%, 5%, and 10% levels, respectively.

	Dependent variable: $S_{j,t}^L - S_{j,t-5}^L$				
	Full Sample (1)	Excluding Tradable Industries (2)	Tradable Industries (3)	Excluding Sectors with Tax-Exempt Firms (4)	Excluding R&D Industries (5)
$C_{j,t}^{(4)} - C_{j,t-5}^{(4)}$	-0.113 ^{***} (0.029)	-0.131 ^{***} (0.033)	-0.036 [*] (0.022)	-0.119 ^{***} (0.029)	-0.125 ^{***} (0.031)
Year FE	Yes	Yes	Yes	Yes	Yes
R^2 (Within)	0.07	0.08	0.02	0.08	0.09
Observations	2,224	1,503	721	2,008	2,011
Observed decline	-0.81	-0.48	-2.30	-0.75	-0.72
Predicted decline	-0.84	-1.04	-0.18	-0.93	-1.01

Table VI: Subsamples. The concentration result is strongest outside tradable industries and robust to key exclusions.

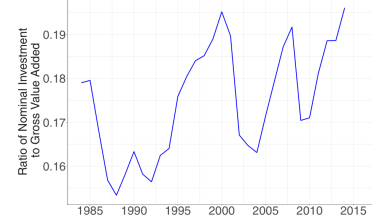


Figure 6. Ratio of investment to gross value added. This figure shows the ratio of investment to gross value added for the U.S. nonfinancial corporate sector over the period 1984 to 2014. See Section VI.A.2 for further details. (Color figure can be viewed at wileyonlinelibrary.com)

Figure 6: Investment share. Investment relative to value added does not show the decline that would be implied by constant-user-cost methods.

8 Robustness and alternative explanations

8.1 Omitted intangible capital

The most important challenge is that measured pure profits might actually be returns to omitted intangible capital, such as brands, organizational capital, employee training, or customer relationships. Barkai addresses this in two ways. First, he adds existing measures of economic competencies and finds that they explain only a small fraction of the pure-profit increase. Second, he runs many hypothetical missing-capital scenarios and finds that explaining the pure-profit rise requires a very large stock of missing intangible capital, often larger than all BEA-measured capital.

8.2 Equity risk premium

If the equity risk premium rose over time, then the required return on equity capital would be understated in later years, and pure profits would be overstated. Barkai studies this explicitly. A large upward trend in the equity premium weakens the increase in pure profits, but the literature generally finds that the equity premium has been flat or declining since the 1980s. Thus, if anything, the baseline may understate the increase in pure profits.

8.3 Depreciation and capital inflation

The capital share depends on depreciation and expected asset-price inflation. Barkai shows that broad changes in these assumptions do not overturn the results. The main reason is that the decline in WACC is large and persistent, while plausible depreciation and inflation adjustments are too small to eliminate the trend.

8.4 Labor income in disguise

Some private business income may be labor income misreported as profits, especially for tax reasons. Barkai discusses evidence from Smith et al. (2019) suggesting that this may overstate the decline in the labor share and the increase in pure profits by about 1.2 percentage points. This is meaningful but much smaller than the 13.5 percentage point rise in the pure profit share.

8.5 Entry costs

The paper's pure profit measure is gross of some entry costs that are not captured as labor costs or investment. Rising entry costs could explain why pure profits rise without entry eliminating rents. Barkai treats this as compatible with the declining competition story rather than as an alternative to it: higher entry costs are one mechanism through which competition can decline.

9 Economic interpretation

9.1 What would the capital-substitution story predict?

If falling labor shares were mainly due to firms substituting capital for labor, then one should observe a rising capital share. A lower required return should lead to enough capital accumulation that capital payments as a share of output rise. Barkai's data show the opposite.

9.2 What does the pure-profit story imply?

If firms gain market power, they can charge higher markups, restrict output and input demand, and earn profits above the cost of labor and capital. In this case both labor and capital shares can fall together. This pattern matches the aggregate evidence and the industry concentration evidence.

9.3 Relation to markups and superstar firms

The results are closely related to the literature on rising markups and superstar firms. A rise in pure profits is macro evidence that firms are capturing rents. The industry results are consistent with the idea that concentration and reallocation toward dominant firms lower labor's share of revenue.

10 Limitations

- **Pure profits are residual:** Any measurement error in gross value added, labor compensation, taxes, capital stocks, or required returns affects the residual.
- **Industry regressions are not causal:** The concentration-labor-share relation is consistent with market power but does not by itself prove that concentration caused the labor-share decline.
- **Intangibles remain hard to measure:** Barkai provides strong scenario evidence, but no paper can fully rule out all forms of missing capital.
- **Equity cost of capital is model-based:** Debt costs are observed more directly; equity costs require assumptions about the equity premium.

- **Aggregate sector focus:** The main accounting is for U.S. nonfinancial corporations, not the whole economy or global economy.

11 Bottom line

The paper's central empirical point is simple but powerful: labor's share fell, capital's share also fell, and pure profits rose. This pattern is difficult to reconcile with a pure labor-to-capital substitution story and is much more consistent with rising market power. The paper's most important contribution is to measure capital costs directly and show that the nonlabor share should not automatically be interpreted as capital income.